COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S1
DATE PERFORMED:	07/16/26	**DATE SUBMITTED:**	
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ACTIVITY NO. 2.1 **Hands-on Activity 2.1 Arrays, Pointers and Dynamic Memory Allocation**

1. PROCEDURE OUTPUT

ILO A: Implement static and dynamic memory allocation & ILO B: Create dynamically allocated objects using pointers and arrays

INPUT

```cpp
#include <iostream> // Table 2-1.
#include <string>

class Student {
    private:
        std::string studentName;
        int studentAge;
    public:
        // Constructor
        Student(std::string newName = "John Doe", int newAge = 18) {
            studentName = std::move(newName);
            studentAge = newAge;
            std::cout << "Constructor Called." << std::endl;
        }

        // Deconstructor
        ~Student() {
            std::cout << "Destructor Called." << std::endl;
        }

        // Copy Constructor
        Student(const Student &copyStudent) {
            std::cout << "Copy Constructor Called" << std::endl;
            studentName = copyStudent.studentName;
            studentAge = copyStudent.studentAge;
        }
```

```
        // Display Attributes
        void printDetails() {
            std::cout << this->studentName << " " << this->studentAge <<
std::endl;
        }

};

int main() {
    Student student1("Roman", 28);
    Student student2(student1);
    Student student3;
    student3 = student2;

    return 0;
}
```

For Table 2-1.

```
#include <iostream> // Table 2-2.
#include <string>

class Student {
    private:
        std::string studentName;
        int studentAge;
    public:
        // Constructor
        Student(std::string newName = "John Doe", int newAge = 18) {
            studentName = std::move(newName);
            studentAge = newAge;
            std::cout << "Constructor Called." << std::endl;
        }

        // Deconstructor
        ~Student() {
            std::cout << "Destructor Called." << std::endl;
        }

        // Copy Constructor
        Student(const Student &copyStudent) {
```

```
            std::cout << "Copy Constructor Called" << std::endl;
            studentName = copyStudent.studentName;
            studentAge = copyStudent.studentAge;
        }

        // Display Attributes
        void printDetails() {
            std::cout << this->studentName << " " << this->studentAge <<
std::endl;
        }

};

int main() {
    const size_t j = 5;

    Student studentList[j] = {};
    std::string namesList[j] = {"Carly", "Freddy", "Sam", "Zack", "Cody"};
    int ageList[j] = {15, 16, 18, 19, 16};

    return 0;
}
```

For Table 2-2.

OUTPUT	

Screenshot	`PS C:\Users\TIPQC\Documents\CPE 010 - CPE21S1 (REYES)> & 'c:\U`
	`-stdout=Microsoft-MIEngine-Out-w1gu1hep.uw1' '--stderr=Microsof`
	`Constructor Called.`
	`Copy Constructor Called`
	`Constructor Called.`
	`Destructor Called.`
	`Destructor Called.`
	`Destructor Called.`
	`PS C:\Users\TIPQC\Documents\CPE 010 - CPE21S1 (REYES)>`

Observation	The program begins with including iostream (input/output stream) and strings, then the class defining the constructor and the big three (that being deconstructor, copy constructor, and copy assignment operator. Once all of those have been defined, the main function is ran to show the static and dynamic allocation, therefore the output.

Table 2-1. Initial Driver Program

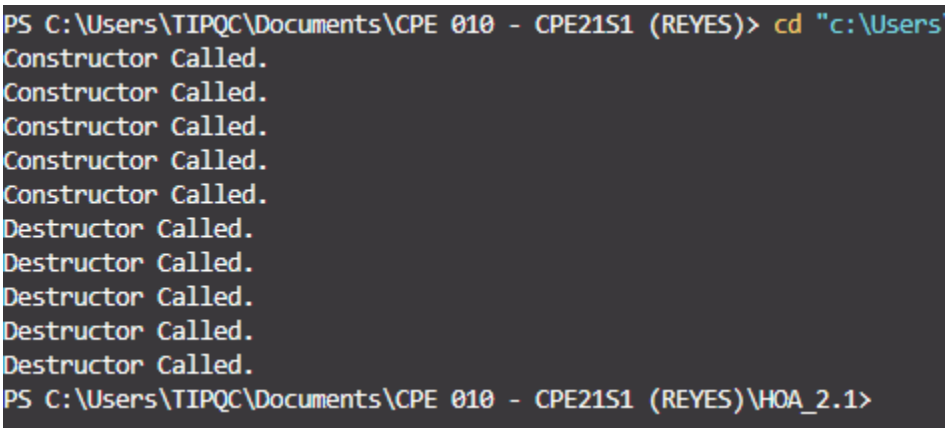
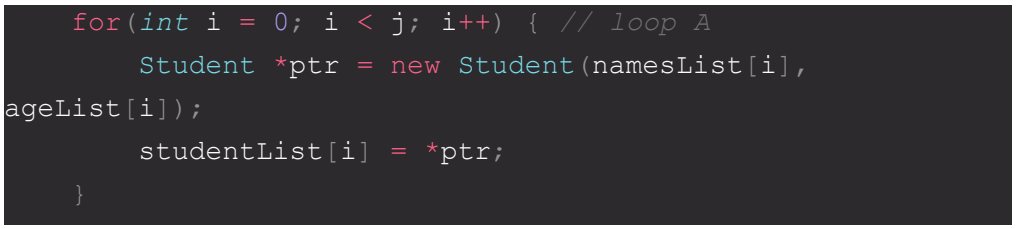
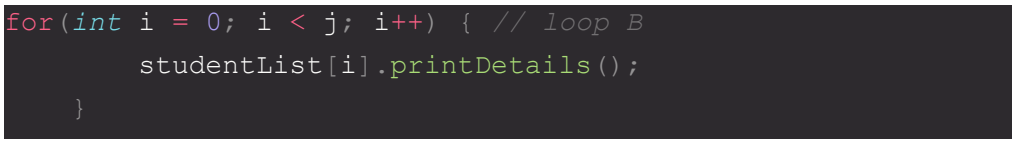
Screenshot	
Observation	Table 2-2 being a modified version of 2-1 giving it an array size of 5, array for student objects containing their names and age. 5 outputs are then seen matching the size of the array, therefore the list being processed.

Table 2-2. Modified Driver Program with Student Lists

Loop A	
Observation	Loop A is responsible for printing out each name and age on the list. If not for this, the output would just be “John Doe” and its set age as 18.
Loop B	
Observation	Loop B prints out the list of names using the Constructor function to display them, relying on Loop A for the format.

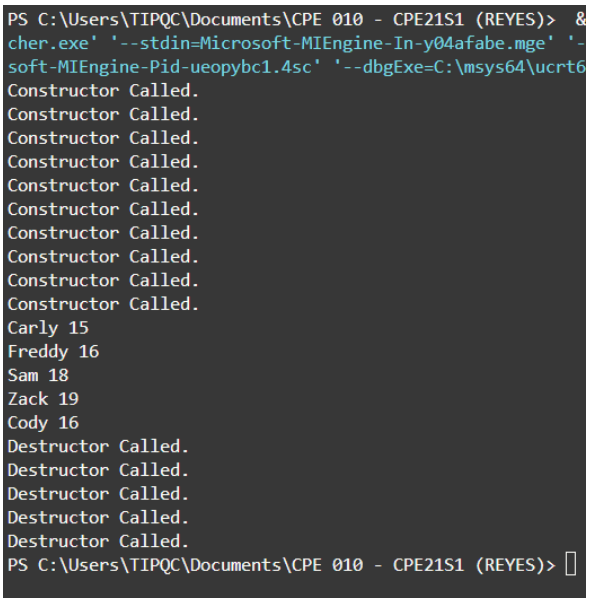
Output	
Observation	The functions inside the class kept executing due to loop B constantly calling them, resulting in the chain of Constructor and Destructor being printed.

Table 2-3. Final Driver Program

Modifications	

```cpp
#include <iostream>
#include <string>

class Student {
    private:
        std::string studentName;
        int studentAge;
    public:
        // Constructor
        Student(std::string newName = "John Doe", int newAge =
18) {
            studentName = std::move(newName);
            studentAge = newAge;
            std::cout << "Constructor Called for " <<
studentName << std::endl;
        }

        // Deconstructor
        ~Student() {
```

```
            std::cout << "Destructor Called for " << studentName
<< std::endl;
        }

        // Copy Constructor
        Student(const Student &copyStudent) {
            std::cout << "Copy Constructor Called" << std::endl;
            studentName = copyStudent.studentName;
            studentAge = copyStudent.studentAge;
        }

        // Display Attributes
        void printDetails() {
            std::cout << this->studentName << " " <<
this->studentAge << std::endl;
        }

};

int main() {
    const size_t j = 5;

    Student studentList[j] = {}; // This is what causes the extra
5 Constructors to pop up on output.
    std::string namesList[j] = {"Carly", "Freddy", "Sam",
"Zack", "Cody"};
    int ageList[j] = {15, 16, 18, 19, 16};

    for(int i = 0; i < j; i++) { // loop A
        Student *ptr = new Student(namesList[i], ageList[i]);
        studentList[i] = *ptr;
        delete ptr; //
    }

    std::cout << std::endl; // Seperating the list of names from
the constructor and destructor

    for(int i = 0; i < j; i++) { // loop B
        studentList[i].printDetails();
    }
```

	``
	    std::cout << std::endl;
	
	    return 0;
	}
	``

Observation	// To be continued

Table 2-4. Modifications/Corrections Necessary

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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